

5 FastAPI Projects for Beginners

From Zero to Job-Ready Backend Developer

These 5 projects are designed for beginners starting their FastAPI journey. Each project builds on the previous one, gradually introducing new concepts. Complete them in order to become a confident Python backend developer.

1. Todo List API

Description

A simple task management API where users can create, view, update, and delete their todos. This is the perfect first project to understand how APIs work, how to connect to a database, and how to structure a FastAPI application.

Key Features

- Create a new todo with title and description
- View all todos or a single todo by ID
- Update todo title, description, or status
- Delete a todo
- Mark todo as complete or incomplete
- Filter todos by status (pending/completed)

Learning Scope

Concept	What You'll Learn
FastAPI Basics	Routes, path parameters, query parameters
HTTP Methods	GET, POST, PUT, DELETE requests
Pydantic Models	Request/response validation and serialization
SQLite Database	Basic database connection and queries
CRUD Operations	Create, Read, Update, Delete patterns

2. Notes API with User Authentication

Description

A personal notes application where users can register, login, and manage their private notes. Each user can only see their own notes. This project teaches you user authentication and how to protect API routes.

Key Features

- User registration with email and password
- User login and logout
- Create, read, update, delete personal notes
- Each user sees only their own notes
- Search notes by title or content
- Password reset functionality

Learning Scope

Concept	What You'll Learn
User Authentication	Register, login, password hashing with bcrypt
JWT Tokens	Creating and validating access tokens
Protected Routes	Middleware to check authentication
User-Data Relations	Linking notes to specific users (foreign keys)
Basic Search	Simple text search in database

3. Personal Expense Tracker

Description

A personal finance API to track daily expenses and income. Users can categorize transactions, set monthly budgets, and view spending reports. This project introduces working with dates, categories, and basic data analysis.

Key Features

- Add income or expense with amount and category
- Categories: Food, Transport, Shopping, Bills, Salary, etc.
- View transactions by date range (today, this week, this month)
- Monthly budget setting with alerts
- Summary report: total income, expenses, and balance
- Category-wise spending breakdown

Learning Scope

Concept	What You'll Learn
Date Handling	Working with dates, filtering by date ranges
Categories	One-to-many relationships (category to transactions)

Aggregations	SUM, COUNT, GROUP BY for reports
Query Filters	Multiple filters (date, category, type)
Data Validation	Validating amounts, dates, and enums

4. URL Shortener Service

Description

A URL shortening service like bit.ly. Users submit a long URL and get a short code. When someone visits the short URL, they are redirected to the original. This project teaches encoding, redirects, and basic analytics.

Key Features

- Submit long URL, get short code (e.g., abc123)
- Redirect short URL to original URL
- Track click count for each short URL
- Set expiry date for short URLs (optional)
- View your created short URLs and their stats
- Custom short code option

Learning Scope

Concept	What You'll Learn
Redirects	HTTP 301/302 redirects in FastAPI
Unique Code Generation	Random string generation, checking uniqueness
Click Tracking	Counting and storing analytics data
Expiry Logic	Checking dates and handling expired URLs
Input Validation	Validating URL format before saving

5. Blog API with Comments

Description

A blogging platform API where users can write posts, add tags, and other users can comment on posts. This is the capstone project combining everything you learned: authentication, relationships, and more complex queries.

Key Features

- User registration and authentication

- Create, edit, delete blog posts
- Add tags to posts (Python, FastAPI, Tutorial, etc.)
- Comment on posts (authenticated users only)
- Like/unlike posts
- View posts by tag or author
- Pagination for posts list

Learning Scope

Concept	What You'll Learn
Many-to-Many Relations	Posts and Tags relationship with junction table
Nested Data	Returning posts with comments and author info
Pagination	Limit, offset, and page-based pagination
Likes System	Toggle like/unlike, counting likes
Query Optimization	Eager loading to avoid N+1 queries

Project Progression Summary

Complete projects in this order for best learning:

#	Project	Level	New Skills
1	Todo List API	Beginner	CRUD, Routes, Pydantic, SQLite basics
2	Notes API	Beginner+	JWT Auth, Password Hashing, Protected Routes
3	Expense Tracker	Intermediate	Date Filters, Aggregations, Reports
4	URL Shortener	Intermediate	Redirects, Code Generation, Analytics
5	Blog API	Intermediate+	Many-to-Many, Pagination, Nested Data

Recommended Tech Stack

- **Framework:** FastAPI
- **Database:** SQLite (beginner) → PostgreSQL (later)
- **ORM:** SQLAlchemy
- **Validation:** Pydantic

- **Auth:** python-jose (JWT), passlib (password hashing)
- **Testing:** Pytest